

Aliah University
End-Semester Examination- 2022
(BTech CSE 2nd Semester)

Paper Name: Programming for Problem Solving
Paper Code: CSEUGES01

Full Marks: 80
Time: 3 Hrs

Group - A

(10X1=10)

1. Fill in the blanks-

- a) C compiler follows _____ approach.
- b) A header file keeps _____ of library functions.
- c) '2xy' is valid variable name in C compiler. _____ [True/ False]
- d) A declaration: float a, b; occupies _____ Bytes of memory.
- e) Operators in C have precedence and it determines which operator is _____.
- f) An instruction: a*=3; means a = _____.
- g) Sscanf() can't read _____ string, gets() can do it.
- h) IDE provides _____, _____ and _____.
- i) What is the answer of: -5%-3? _____
- j) A control character for an address is %_____

Group - B

(Answer any 6 questions)

(6X5=30)

2. Mention differences between pre-increment and post-increment with suitable example.
3. Differentiate 'auto' and 'static' storage classes with examples.
4. Explain the basic string standard functions with proper examples.
5. How while() and do-while() loops are different from one another? Explain with example.
6. Write a C program to insert some integer values in an Array and then print the average of them.
7. Write a program in C to find out maximum and minimum number out of three integer values, by using conditional operator.
8. Write down outputs for the programs below: (give explanation)

a)

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int i,j;
    for(i=1;i<4;i++)
        for(j=1;j<3;j++)
        {
            if(i<=j)
                break;
            printf("%d %d\n",i,j);
        }
    return 0;
}
```

b)

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int i,j;
    for(i=1;i<4;i++)
        for(j=1;j<3;j++)
        {
            if(i==j)
                continue;
            printf("%d %d\n",i,j);
        }
    return 0;
}
```

a=1 b=4 c=9

a=9 b=7, c=808

max=789

min=444

Group - C
(Answer any 4 questions)

(10X4=40)

9.

- a) Define Recursion. Write a program in C to find out Greatest Common Divisor between two integer values by using recursion. (5*2=10)
b) Write a C program to print the following pattern:

```
1
1 2 3
1 2 3 4 5
```

10.

- a) Write a C program to check whether a number is palindrome or not. (4+1+2+3=10)
b) Define function. What are the advantages and disadvantages of using function? Give solution to overcome the disadvantages of using a function. Explain with example.

11.

- a) Discuss output differences between "Call by value" and "Call by reference" for swapping two integer numbers. (5*2=10)
b) Write a program in C to sort the array (mentioned below) in ascending order -

34	23	45	4	67	17	7
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12.

- a) Write a note on - Switch and case. What is the role of 'break' in switch-case? Explain with example.
b) Write a program in C to add first 'n' prime numbers and print the result. (Use separate function for prime number checking) (5*2=10)

13.

- a) Define Pointer. Write a program in C to get reverse of a string, using pointers. (2+5+3=10)
b) Write short notes on - relational operators and logical operators in C programming.

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a = 4 b = 7 c = 5

max = 0 min = 4

max = 7

min = 3